Enucleation versus preservation of blind eyes following plaque radiotherapy for choroidal melanoma

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ABSTRACT

Background: Currently available information about patients with posterior uveal melanoma treated by plaque radiotherapy is insufficient to determine what to do about eyes that become blind as a consequence of the tumour and its treatment. Should they be enucleated, or is ocular preservation just as good in terms of survival?

Methods: We performed a retrospective survival analysis of secondary enucleation versus ocular preservation in patients with a posterior uveal melanoma treated by plaque radiotherapy whose irradiated eye became completely blind following that treatment. Of the 79 patients who fulfilled defined inclusion criteria, 25 underwent secondary enucleation of the blind eye, and 54 retained their irradiated blind eye.

Results: Most of the baseline demographic and tumour-related variables evaluated were similarly distributed between the subgroups. The 5-year, 10-year and 15-year all-cause death rates in the secondary enucleation subgroup were 24.7%, 51.5% and 52.0% respectively, and those in the ocular preservation subgroup were 7.4%, 32.9% and 48.1% respectively. In spite of the apparent slight difference between the curves, the difference was not statistically significant (p = 0.41, Mantel–Haenszel test).

Interpretation: Although a retrospective study of this type has several limitations, our results suggest that secondary enucleation is not likely to substantially improve survival of patients whose irradiated eye becomes totally blind following plaque radiotherapy for choroidal or ciliochoroidal melanoma.

Currently available survival data for patients with a primary posterior uveal melanoma treated by plaque radiotherapy are inadequate to determine the advisability of enucleating or retaining the irradiated eye if it becomes blind subsequent to the treatment. Some experienced clinicians believe that secondary enucleation of such eyes is likely to improve the patient's survival prognosis and should therefore be recommended uniformly. With equal conviction, other experienced clinicians believe that enucleation of such eyes is unlikely to improve survival and should therefore not be recommended unless other factors (such as severe eye pain or intraocular tumour relapse[1]) prompt this surgery.

We attempted to address this issue by means of a retrospective survival study. The principal objective

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